Name _____

# Activity: Reaction Time

In this activity we will look at the effect of various conditions on reaction time. It is difficult to calculate reaction time based on time (seconds) because the amount of time is so small – it is difficult to measure. Instead, since we know acceleration due to gravity, we can measure how far a meter stick falls before you catch it. From that distance, we can calculate reaction time.

## Materials:

- Meter Stick
- CD/MP3 Player, Radio, or Television
- Your Nervous System

## Procedure:

IMPORTANT: YOU MUST BE VERY QUIET DURING STEPS 1-3. THERE SHOULD BE ABSOLUTELY NO TALKING.

1. Have your lab partner hold the meter stick so that the "0cm" mark is between the thumb and index finger of your dominant hand (the hand you write with). Your thumb and index finger should be held one inch apart. Without warning, your lab partner should drop the meter stick, and you should catch it as soon as possible. Write down how far the meter stick fell in the *Data Collection and Calculations* section of this lab.

2. Repeat the preceding 9 more times (for a total of t10) and calculate the average distance for the 10 trials.

3. Calculate your reaction time using the formula below.

   Reaction Time = $\sqrt{2s / a}$

   $a = 980 cm / sec^2$
   s = average distance (cm)

4. Repeat steps #1-3 with a loud room and with a dark room.

5. Collect class data for all three conditions and record the data in the *Data Collection and Calculations* section of this lab.

6. Make groups (intervals) for reaction times for the class (0 - .05 seconds, 0.06 - 1.0 seconds, etc.) and count the number of people in each group for each condition. Record the totals in the appropriate columns.

7. Make a bar graph with reaction time on the X-axis and frequency on the Y-axis. Plot data for all three conditions.

## Data Collection and Calculations:

| Conditions | Distance Meter Stick Fell (cm) – Trial #: | | | | | | | | | | Sum | Avg. | Reaction Time |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | | | |
| Silent Room | | | | | | | | | | | | | |
| Loud Room | | | | | | | | | | | | | |
| Dark Room | | | | | | | | | | | | | |

| Group Data | | | |
|---|---|---|---|
| Name | Reaction Time | | |
| | Silent | Loud | Dark |
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| Frequency Breakdown | | | |
|---|---|---|---|
| Groups | Reaction Time | | |
| | Silent | Loud | Dark |
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## Activity Wrap-up:

Answer questions and attach your graph (make sure to put your name as part of the title of your graph).

**Short Answer Questions:**

1. List three factors other than noise and light that might affect reaction time.

2. Under what conditions did most people have the fastest reaction times?

**Discussion Questions:**

3. Diagram and label the path taken by the nerve impulse in this lab. Include the following in you diagram (eye, light receptor, motor neuron, interneuron, sensory neuron, spinal cord, brain, and effector, muscle, and hand).

4. What impact would being drunk or high on marijuana have on the results? Explain where these drugs act.

5. Discuss your graph. If you did it correctly, it is probably a bit confusing to look at. Explain what the graph shows.

## Class Data Overhead:

| Name | Reaction Time | | |
|---|---|---|---|
| | Silent | Loud | Dark |
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